

# Demonstrating real-time slow-motion experience through parallel video presentation

Goki Muramoto  
goki.muramoto@gmail.com  
The University of Tokyo  
Japan

Sohei Wakisaka  
wakisaka@kmd.keio.ac.jp  
Keio University  
Japan

Hiroto Saito  
saito@star.rcast.u-tokyo.ac.jp  
The University of Tokyo  
Japan

Masahiko Inami  
inami@star.rcast.u-tokyo.ac.jp  
The University of Tokyo  
Japan

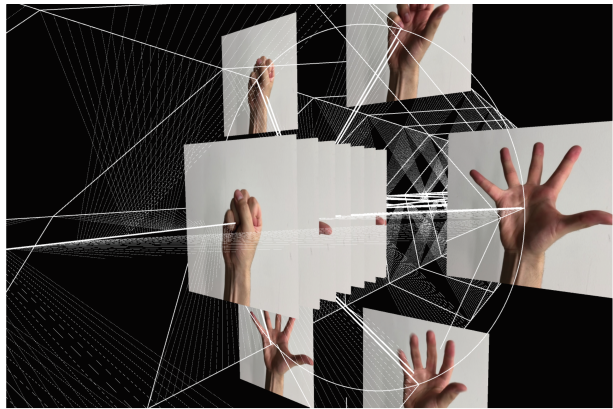
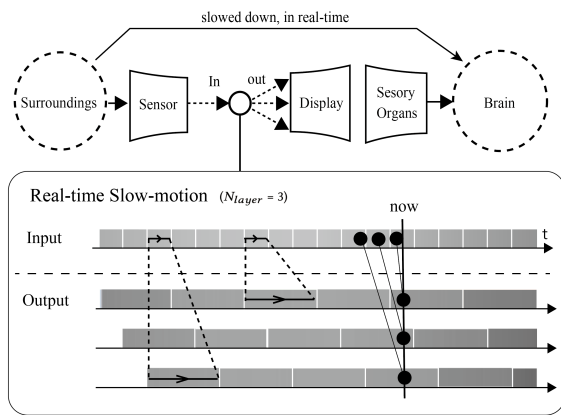


Figure 1: The framework of Real-time Slow-motion and its visualization

## ABSTRACT

Real-time and slow-motion are basically incompatible properties in the technologies of recording and playback. We propose a framework to realize the coexistence of both using parallel video presentation based on the integrative capacity and the temporal nature of cognition. In this framework, camera input is divided into short time durations, distributed into layered timelines, and then stretched out. And then, all layers are composited and presented in a display simultaneously. In our demonstration, users can experience their surroundings through see-through goggles or digital mirrors in which this framework is embedded. Under certain conditions, the perception of the layers is integrated into the cognitive process, resulting in the coexistence of real-time and slow-motion. This research suggests a new field of “Temporal Editing to Humans.”

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## CCS CONCEPTS

• **Human-centered computing** → **Displays and imagers**; • **Hardware** → *Emerging interfaces*.

## KEYWORDS

Time-perception, Real-time, Slow-motion, Sensory augmentation

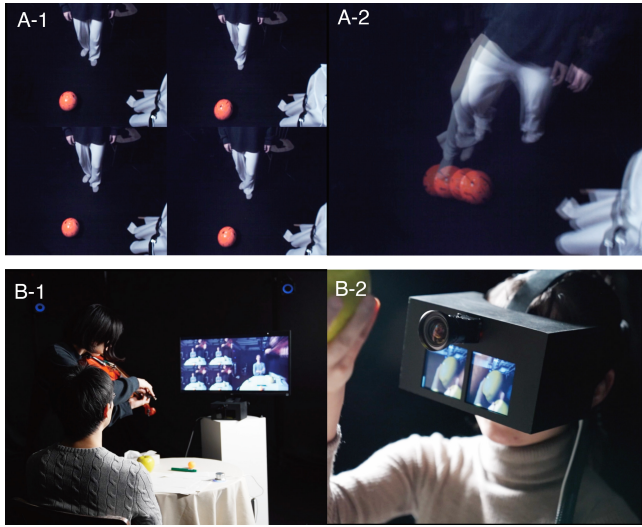
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## 1 INTRODUCTION

Input/output technologies of sensory information, represented by the recording/playback of video images, have expanded human perceptual experience. In particular, real-time editing (e.g. augmented reality) and temporal editing (e.g., slow-motion) have contributed to this expansion. The former can diversify the user’s perceptual experience of their current surroundings or personal activities with the *playback of the present* as its background. The latter can change the user’s perspective on spatio-temporal changes by transforming the time scale and temporal resolution of input information.

However, real-time editing and temporal editing are incompatible because the playback at different speeds of a recorded video will diverge over time. In other words, to achieve *playback of the present*, the time scale of the recorded video cannot be edited.



**Figure 2: A: Two methods of parallel display of the output timeline. B: Two ways to experience this video presentation**

We propose *Real-time Slow-motion* as a framework that coexists real-time and one of the basic temporal editing, slow-motion by adding temporal editing within a range that the user subjectively perceives as the present. In this framework, we attempt to realize slow-motion that does not diverge from real-time by dividing the input within a short duration into multiple layers, stretching them on each layer, and presenting them simultaneously to the user. We have reported that this framework could induce a sense of slowness through psychological experiments using two implementations that limit layer number to two: one in which each layer is assigned to the right and left eye images of the head mounted display (HMD), and the other optional implementation in which the layers are blended into one image by overlaying them [Muramoto et al. 2024].

In this demonstration, we interpreted the framework broadly and implemented that further expands the number of layers to three or more to provide a more enhanced slow-motion experience. Unlike previous approaches, which switch to slow-motion or add slow-motion only when certain events occur [Knierim et al. 2020; Nejime et al. 1996; Orikasa et al. 2010], our approach aims to partially overcome the principle constraints of recording and playback by exploiting the temporal properties of cognition that the subjective present has a width, and that temporal relations are flexible within its width [James’s 1863; Titchener 1908].

## 2 FRAMEWORK

The *Real-time Slow-motion* framework consists of the following processes (see our previous work [Muramoto et al. 2024] for detail):

- (1) *Distribution*: Inputs from the environment are separated on the timeline by a certain duration (SD: Segment Duration) and distributed in sequence on  $N$  parallel layers.
- (2) *Stretching*: The distributed input is stretched to be  $1/N$  times slower in each layer, with the head fixed.
- (3) *Composition*: All layers are juxtaposed and output simultaneously so that each of the layers is equally perceptible. Here,

we expect that visual stimuli from each layer are integrated into a single timeline by the user’s cognitive processes.

## 3 IMPLEMENTATION

The basis of the implementation of *Real-time Slow-motion* consists of a high-frame-rate camera (120FPS, full HD), a display (full HD), and the computer connecting between both. First, the camera image is sent to a buffer on the computer. The buffered images are then edited according to the framework described in the section 2. There are two methods available for displaying the output timeline in parallel: the split screen method and the overlay screen method. The split screen method arranges the images on each layer clockwise (Figure 2 A-1), while the overlay screen method blends the images on each layer with transparency. Additionally, there are two ways to experience this parallel video presentation.

**See-through goggle type** The camera is embedded in the HMD. Users can see various things, such as their bodies, apples, and balls, through these goggles (Figure 2 B-1).

**Mirror type** There are big displays with a front-facing camera. Like a mirror, the camera images reflect back to the participants by the displays. (Figure 2 B-2).

## 4 USER EXPERIENCE

Attendees can see their own bodies and prepared objects through mirrors or see-through goggles. Each moment of movement in the environment appears as slow motion, but since the delay of the image is guaranteed to be within  $0 < d < SD(Layer - Number - 1)$ , it is perceived as a real-time event, resulting in the contradictory experience of real-time and slow motion coexisting. The user can adjust SD with the knob at hand. At lower SD (0.20 msec and below), the user experiences the illusion that the motion appears slow while it is almost real-time. At higher SD, the layers are separated like chronophotography, showing clear motion details as if watching a normal recorded slow-motion, although each layer is experienced as an event of the last few seconds, allowing for rough feedback.

## ACKNOWLEDGMENTS

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